The Physics of Sound Transmission

through Building Partitions

A companion to the prediction engine:
derivations, reasoning, and worked examples

Contents

Preface

This book has one job: to make sure you never trust a number from the software without knowing where it came from. Anyone can read a transmission-loss curve off a screen. The consultant who can *defend* that curve—who knows which physical effect produced each wiggle, which assumption is holding the prediction up, and where the honest uncertainty lies—is doing something quite different. Everything the engine computes is derived here, from the beginning, with every symbol defined and every step shown, and worked by hand at least once so you can follow the reasoning rather than take it on faith.

I have tried to write it the way one physicist explains a subject to another over a table: start from the physical picture, ask what is really going on, and only then reach for the mathematics—and when we do reach for it, do it carefully and completely, because a derivation with a step missing is just an assertion in disguise.

Each chapter carries a status tag: [**Validated**] means the corresponding engine module is implemented and checked against a canonical published result; [**In progress**] and [**Planned**] mean what they say. Nothing is presented as settled until it has passed that test.

A note on symbols. The quantities used throughout are collected here for reference; each is also defined at the point it first appears.

Symbol	Meaning	Units
p	acoustic pressure	Pa
v	normal particle velocity	$\mathrm{m\,s^{-1}}$
ω	angular frequency, $\omega = 2\pi f$	$\mathrm{rad\,s^{-1}}$
ρ_0	density of air	$\mathrm{kg\,m^{-3}}$
c_0	speed of sound in air	$\mathrm{m\,s^{-1}}$
$Z_0 = \rho_0 c_0$	characteristic impedance of air	$\mathrm{Pa\,s\,m^{-1}}$
θ	angle of incidence (from the normal)	rad
$k_0 = \omega/c_0$	acoustic wavenumber	$\mathrm{m^{-1}}$
m	surface (areal) mass density of a leaf	$\mathrm{kg\,m^{-2}}$
$E,\ \nu,\ \eta$	Young's modulus, Poisson's ratio, loss factor	Pa, –, –
h	plate thickness	m
D	bending stiffness	N m
k_B	free bending wavenumber	$\mathrm{m^{-1}}$
f_c	critical (coincidence) frequency	Hz
τ	transmission coefficient	–
σ	radiation efficiency	–
G	shear modulus	Pa
k_z	normal (through-thickness) component of a wavenumber	$\mathrm{m^{-1}}$
u	normal displacement of a leaf	m
b	stud spacing	m
d	cavity (stud) depth	m
C_M	equivalent translational compliance of a line connection	$\mathrm{Pa^{-1}}$
K'	translational line stiffness, $K' = 1/C_M$	$\mathrm{N\,m^{-2}}$

Chapter 1

Fundamentals

[Validated]

1.1 What we are really asking

Picture a wall between you and a busy road. On the far side, air is being shaken by traffic; a little of that shaking leaks through the wall and reaches your ear. The entire subject is the study of how little.

Let W_i be the sound power arriving at the wall and W_t the power that emerges on the other side. Their ratio is the *transmission coefficient*,

$$\tau = \frac{W_\mathrm{t}}{W_\mathrm{i}}, \qquad 0 < \tau \leq 1. \tag{1.1}$$

A perfect wall has $\tau = 0$; an open window has $\tau = 1$. Because τ ranges over many orders of magnitude—a good wall transmits perhaps one part in a million—we work with its logarithm, the *sound transmission loss*,

$$\mathrm{TL} = -10 \log_{10} \tau \quad [\mathrm{dB}]. \tag{1.2}$$

The decibel is just a convenient ruler for ratios: every factor of ten in $1/\tau$ adds 10 dB. Our whole task is to compute τ as a function of frequency and angle; everything else is bookkeeping.

1.2 Sound as pressure and motion

A sound wave is two things happening together: the air is squeezed (a pressure p) and the air moves (a normal velocity v). In a single plane wave travelling through free air these are not independent—they are locked together by the medium. For a wave moving along its direction of travel,

$$p = \rho_0 c_0 \, v, \tag{1.3}$$

where ρ_0 is the density of air and c_0 the speed of sound. The proportionality constant $Z_0 = \rho_0 c_0$ is the *characteristic impedance* of air: the "stiffness" the medium presents to being pushed. (Air at room temperature has $Z_0 \approx 415 \ \mathrm{Pa\,s\,m^{-1}}$.)

Now let the wave strike the wall not head-on but at an angle θ measured from the normal. Only the component of motion *along the normal* pushes on the wall; the along-the-wall component slides past it. The velocity that matters is $v \cos\theta$, so the impedance the wall feels for an oblique wave is larger by $1/\cos\theta$:

$$Z_a(\theta) = \frac{\rho_0 c_0}{\cos\theta}. \tag{1.4}$$

This single factor—the geometry of a wave hitting a surface at a slant—will follow us through the entire book. Throughout we take a time dependence $\mathrm{e}^{\mathrm{i}\omega t}$, so a time derivative becomes multiplication by $\mathrm{i}\omega$.

1.3 The simplest wall: a limp mass

Strip a wall down to its barest idealisation: a thin sheet of mass, so floppy it has no stiffness of its own—a "limp mass" of areal density m (kilograms per square metre). What does it do to a sound wave?

It obeys Newton's second law. The net force per unit area on the sheet is the pressure difference across it, $p_1 - p_2$ (front minus back), and this accelerates its mass:

$$m\,\frac{\partial v}{\partial t} = p_1 - p_2 \quad\Longrightarrow\quad \mathrm{i}\omega m\, v = p_1 - p_2, \tag{1.5}$$

using $\partial/\partial t \to \mathrm{i}\omega$. The sheet is thin, so the air's velocity is continuous through it: $v_1 = v_2 = v$. Collecting these two facts,

$$p_1 = p_2 + \mathrm{i}\omega m\, v, \qquad v_1 = v_2, \tag{1.6}$$

which we write compactly as a matrix relating the (pressure, velocity) pair on the front face to the pair on the back:

$$\begin{pmatrix} p_1 \\ v_1 \end{pmatrix} = \begin{pmatrix} 1 & \mathrm{i}\omega m \\ 0 & 1 \end{pmatrix} \begin{pmatrix} p_2 \\ v_2 \end{pmatrix}. \tag{1.7}$$

This little 2×2 object is the *transfer matrix* of the layer. It is the atom from which every build-up in this book is assembled. The quantity $\mathrm{i}\omega m$ is the layer's impedance: heavy walls and high frequencies both make it large, which is the mass law in embryo.

1.4 Stacking layers: the transfer matrix method

A real partition is a stack of such atoms—plasterboard, an air gap, insulation, more plasterboard. Each layer k has its own transfer matrix $\mathbf{T}_k(\omega, \theta)$. Because the back face of one layer is the front face of the next, the whole partition is simply the ordered product

$$\mathbf{T} = \mathbf{T}_1\,\mathbf{T}_2 \cdots \mathbf{T}_N = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \tag{1.8}$$

There is no special case for "double wall" or "triple glazing"; those are just different products of the same atoms. That is the whole reason this method can be trusted on a build-up nobody has ever measured.

To get the transmission coefficient we attach air to both sides. On the incident side the field is an incoming wave plus a reflected one; on the far side there is only an outgoing (transmitted) wave. Writing the transmitted pressure as p_t, the far-side relations are

$$p_2 = p_\mathrm{t}, \qquad v_2 = \frac{p_\mathrm{t}}{Z_a}, \tag{1.9}$$

since a lone outgoing wave obeys $p = Z_a v$. Push these through the matrix:

$$p_1 = Ap_2 + Bv_2 = \Big(A + \tfrac{B}{Z_a}\Big)p_\mathrm{t}, \tag{1.10}$$

$$v_1 = Cp_2 + Dv_2 = \Big(C + \tfrac{D}{Z_a}\Big)p_\mathrm{t}. \tag{1.11}$$

On the incident side the incoming pressure p_i can be recovered from the combination $p_1 + Z_a v_1$ (the reflected part cancels), giving $2p_\mathrm{i} = p_1 + Z_a v_1$. Substituting,

$$2p_\mathrm{i} = \Big(A + \tfrac{B}{Z_a} + Z_a C + D\Big)p_\mathrm{t}. \tag{1.12}$$

Since for equal media on both sides $\tau = |p_{\mathrm{t}}/p_{\mathrm{i}}|^2$, we arrive at the master formula for the oblique-incidence transmission coefficient of *any* layered wall:

$$\boxed{\tau(\theta) = \left| \frac{2}{A + B/Z_a + Z_a C + D} \right|^2} \tag{1.13}$$

[Engine module M2, validated: the assembler reproduces the analytic single- and double-leaf results identically.]

1.5 Sound from all directions: the diffuse field

In a room the sound does not arrive from one angle but from all of them at once. We must average $\tau(\theta)$ over a hemisphere of incidence. Two geometrical weights enter. First, a patch of wall intercepts power from a given direction in proportion to its *projected* area, a factor $\cos\theta$. Second, the amount of "sky" at angle θ—the solid angle in a ring at that tilt—is proportional to $\sin\theta$. Hence the random-incidence (diffuse) transmission coefficient is the power-weighted average

$$\tau_{\mathrm{d}} = \frac{\displaystyle\int_0^{\theta_{\mathrm{lim}}} \tau(\theta)\, \cos\theta\, \sin\theta\, \mathrm{d}\theta}{\displaystyle\int_0^{\theta_{\mathrm{lim}}} \cos\theta\, \sin\theta\, \mathrm{d}\theta}, \tag{1.14}$$

known as Paris' formula. The upper limit θ_{lim} is *not* $\pi/2$: a real, finite wall does not radiate at true grazing incidence, and Chapter 3 shows why the honest limit is about 78°.

1.6 One number: the ratings

Finally the one-third-octave TL spectrum is condensed to a single figure by sliding a standard reference contour against it until a rule on unfavourable deviations is satisfied: this yields the weighted sound reduction index R_{w} (with spectrum-adaptation terms C and C_{tr}) under ISO 717-1, and the sound transmission class STC under ASTM E413. The term C_{tr} is deliberately weighted toward low frequencies because it represents *road-traffic* noise—a point we will cash in when we make predictions audible in Chapter 9.

Chapter 2

The single leaf

[Validated]

2.1 Why a real plate is not limp

The limp mass of Chapter 1 is honest only at low frequencies. A real board is *stiff*: bend it and it pushes back. That stiffness lets it carry bending waves along its own surface, and those waves are the source of the single most important feature of any real wall—the coincidence dip.

2.2 Bending waves and stiffness

A thin plate in free flexure obeys

$$D\,\nabla^4 w + m\,\frac{\partial^2 w}{\partial t^2} = 0, \tag{2.1}$$

where w is the out-of-plane displacement and D is the *bending stiffness*

$$D = \frac{Eh^3}{12\,(1-\nu^2)}, \tag{2.2}$$

with E Young's modulus, h the thickness and ν Poisson's ratio. (The h^3 is why stiffness grows so fast with thickness, and the $1-\nu^2$ is the plate's resistance to the sideways bulging that accompanies bending.) Seeking a travelling-wave solution $w \propto \mathrm{e}^{\mathrm{i}(kx-\omega t)}$ turns the differential equation into an algebraic one, $Dk^4 - m\omega^2 = 0$, whose root is the free *bending wavenumber*

$$k_{\mathrm{B}} = \left(\frac{m\omega^2}{D}\right)^{1/4}. \tag{2.3}$$

Notice it grows only as $\sqrt{\omega}$: bending waves get shorter with frequency, but slowly. Keep that in mind—in Chapter 4 it will get us into trouble and we will have to fix it.

2.3 The impedance of a stiff plate

Now drive the plate not by shaking it uniformly but with a sound wave that runs along its surface with *trace wavenumber* $k_t = k_0 \sin\theta$ (the imprint the oblique acoustic wave leaves on the wall). The plate's equation of motion at that spatial frequency gives a pressure-to-velocity ratio—its impedance. Allowing the stiffness to be slightly lossy, $D \to D(1+\mathrm{i}\eta)$ with η the loss factor,

$$Z_p(\omega,\theta) = \frac{D(1+\mathrm{i}\eta)k_t^4 - m\omega^2}{\mathrm{i}\omega} = \mathrm{i}\omega m\left[1-(1+\mathrm{i}\eta)\left(\tfrac{k_t}{k_{\mathrm{B}}}\right)^4\right], \tag{2.4}$$

where the last step used $k_{\mathrm{B}}^4 = m\omega^2/D$. Two limits check it: when the wall is stiff or the frequency low, $k_t \ll k_{\mathrm{B}}$ and the bracket is ≈ 1, recovering the limp mass $\mathrm{i}\omega m$ of Eq. (1.7); when $k_t \to k_{\mathrm{B}}$ the bracket collapses.

2.4 Coincidence

That collapse is the heart of the matter. Define the *critical frequency* f_{c} as the frequency at which a free bending wave and a grazing acoustic wave have the same wavelength, $k_{\mathrm{B}} = k_0$:

$$\left(\tfrac{m\omega_{\mathrm{c}}^2}{D}\right)^{1/4} = \frac{\omega_{\mathrm{c}}}{c_0} \implies f_{\mathrm{c}} = \frac{c_0^2}{2\pi}\sqrt{\frac{m}{D}}. \tag{2.5}$$

A short algebraic tidy-up rewrites the awkward ratio in Eq. (2.4) in terms of it,

$$\left(\tfrac{k_t}{k_{\mathrm{B}}}\right)^4 = \left(\tfrac{f}{f_{\mathrm{c}}}\right)^2 \sin^4\theta, \tag{2.6}$$

so that at $f = f_{\mathrm{c}}$ there is always some angle ($\sin^2\theta = f_{\mathrm{c}}/f$, here $\theta = 90^\circ$) for which the bracket vanishes, the impedance drops to only its small damping part, and the wall goes acoustically transparent. This is *coincidence*: the incident wave and the plate's own bending wave march in step, the plate offers no resistance, and TL plunges. Above f_{c} the wall recovers, its performance now set by how much damping η it has.

2.5 Worked example: where does the dip fall?

Take 13 mm plasterboard of the NRCC "type A" variety, whose Young's modulus has been *measured*: $E = 1.70$ GPa (Hirakawa & Davy, 2015). With $\rho = 770\ \mathrm{kg\,m^{-3}}$ and $\nu = 0.3$, the areal mass is $m = 0.013 \times 770 = 10.0\ \mathrm{kg\,m^{-2}}$ and the bending stiffness is

$$D = \frac{Eh^3}{12(1-\nu^2)} = \frac{1.70 \times 10^9 \times (0.013)^3}{12 \times 0.91} \approx 342\ \mathrm{N\,m}, \qquad f_{\mathrm{c}} = \frac{c_0^2}{2\pi}\sqrt{\frac{m}{D}} \approx 3.2\ \mathrm{kHz}. \tag{2.7}$$

The engine puts the dip there. Two remarks on provenance, because they mattered in validation (Chapter 6). First, an older calibration—Davy's empirical product $mf_{\mathrm{c}} = 31000\ \mathrm{kg\,m^{-2}\,s^{-1}}$, fitted to sound-insulation data—gives 3100 Hz for this board: for 13 mm boards the two anchors agree. For the stiffer 16 mm boards they do not: the measured moduli (2.2–2.4 GPa) place f_{c} near 2.1 kHz, whereas the fitted product implies 2.8 kHz, and the validation against measured walls decided firmly in favour of the measured moduli. When a fitted constant and a measured constant disagree, trust the measurement. Second, the *same* formula, fed 8 mm float glass, returns ≈ 1700 Hz, and 6 mm glass ≈ 2500 Hz—no material-specific tuning anywhere, just the physics of Eq. (2.4).

Chapter 3

Finite size

[Validated] — the flagship result

3.1 The lie of the infinite panel

Everything so far quietly assumed the wall is infinitely wide. That assumption tells a specific lie: an infinite panel, below its critical frequency, would radiate perfectly at grazing incidence. Real, finite walls do not. The consequence is that infinite-panel theory *under*-predicts TL below f_c, exactly the region that matters most for lightweight construction. No serious tool can leave this uncorrected—and the simplified ones paper over it with an empirical fudge (a fictitious limiting angle). We will do it from the physics instead.

3.2 Radiation efficiency

The honest quantity is the *radiation efficiency* σ: how effectively a vibrating panel of finite size $L_x \times L_y$ actually launches sound into the air, compared with an ideal piston. For an infinite panel carrying a supersonic trace wave $\sigma = 1/\cos\theta$; for a finite one it is smaller below coincidence, because the panel's edges let radiation "leak away" rather than build up. The finite transmitted coefficient is the infinite one re-weighted by this true efficiency,

$$\tau_{\text{finite}}(\theta) = \tau_\infty(\theta)\, \sigma(\theta)\, \cos\theta, \tag{3.1}$$

applied to the *transmitted* power only, with $\theta_{\text{lim}} = 78°$.

3.3 Spatial windowing

To compute σ we take the infinite panel's vibration and multiply it by the panel's finite aperture—a spatial window—then ask how much of the resulting wavenumber content lies inside the "radiation circle" $|\mathbf{k}| < k_0$ that can couple to real sound. Villot's result for a structural wave of wavenumber k_p travelling at heading τ across an $L_x \times L_y$ baffled panel is

$$\sigma(k_p, \tau) = \frac{S}{\pi^2} \iint\limits_{k_r<k_0} W_x\, W_y\, \frac{k_0}{\sqrt{k_0^2 - k_r^2}}\; \mathrm{d}k_r\; \mathrm{d}\varphi, \tag{3.2}$$

where $S = L_x L_y$ is the panel area, k_r and φ are polar coordinates in the radiating-wavenumber plane, and W_x, W_y are the (sinc-squared) Fourier transforms of the finite aperture in each direction, $W_x = \left[1 - \cos\left((k_r \cos\varphi - k_{px})L_x\right)\right] / \left[(k_r \cos\varphi - k_{px})L_x\right]^2$, with $k_{px} = k_p \cos\tau$ (and similarly for y). The factor $k_0/\sqrt{k_0^2 - k_r^2}$ is the radiation impedance of air for each emitted component; the substitution $k_r = k_0 \sin\psi$ tames its grazing singularity cleanly.

3.4 A test that needs no figure

Here is why we can trust it. Let the panel grow without bound, $L_x, L_y \to \infty$. The window functions W_x, W_y become ever narrower and, in the limit, act as Dirac deltas that pick out exactly the trace wavenumber. Carrying the delta through the integral collapses Eq. (3.2) to $\sigma = k_0/\sqrt{k_0^2 - k_p^2}$, and with $k_p = k_0 \sin\theta$ this is precisely $1/\cos\theta$—so $\sigma \cos\theta \to 1$, the infinite panel, *exactly.* The engine reproduces this to four decimal places. A correctness test that appeals only to a known analytic limit, not to a fitted curve, is the strongest kind we can have.

3.5 Worked example: the 3 mm aluminium plate

Aluminium has $\rho = 2700$ kg m^{-3}, $E = 69$ GPa, $\nu = 0.3$. A 3 mm, 1×1 m plate therefore has $m = 8.1$ kg m^{-2}, $D = 170.6$ N m, and $f_c = 4080$ Hz. Evaluating the radiation efficiency at the bending wavenumber across frequency gives, in decibels $10 \log_{10} \sigma$:

f (Hz)	$10 \log_{10} \sigma$ (dB)
100	−26 (small panel radiates weakly)
1600	−17
4080	+5 (peak, exactly at f_c)
8000	+1.5 (settling toward 0 above coincidence)

These reproduce the published curve of Rhazi & Atalla (2010, Fig. 1). Because this is the engine's *own* radiation efficiency, verified against a canonical result, the finite-size lift it applies to every other panel is trustworthy by construction. *[Engine module M1, validation gate passed.]*

3.6 What it does to a real wall

For 13 mm gypsum at the standard 3.05×2.44 m laboratory size, the finite-size correction adds +3.2 dB at 100 Hz, +0.8 dB at 500 Hz, and essentially nothing above 2 kHz—largest where the acoustic wavelength approaches the panel size, vanishing where the panel looks effectively infinite. It emerges from Eq. (3.2); we never chose it.

3.7 Where the windowing formula comes from

We wrote Eq. (3.2) down; here is where it comes from, because a formula you cannot derive is one you cannot defend. Nothing below is assumed beyond the Rayleigh integral and the finite size of the panel.

A baffled planar surface in the plane $z = 0$, vibrating with normal velocity $v(x, y)\, e^{i\omega t}$, radiates into the half-space $z > 0$ a pressure given by the first Rayleigh integral,

$$p(\mathbf{r}) = \frac{i\omega\rho_0}{2\pi} \iint_S v(x_0, y_0) \frac{e^{-ik_0 R}}{R} \, dx_0 \, dy_0, \qquad R = |\mathbf{r} - \mathbf{r}_0|. \tag{3.3}$$

The time-averaged radiated power is the integral of the normal acoustic intensity over the surface,

$$W = \tfrac{1}{2} \operatorname{Re} \iint_S p(x, y, 0)\, v^*(x, y) \, dx \, dy. \tag{3.4}$$

The step that unlocks everything is to move to the spatial-wavenumber domain. Define the two-dimensional Fourier transform $\tilde{v}(k_x, k_y) = \iint v(x, y)\, e^{i(k_x x + k_y y)} \, dx \, dy$. By Parseval's theorem,

$$W = \frac{1}{2(2\pi)^2} \operatorname{Re} \iint \tilde{p}(k_x, k_y)\, \tilde{v}^*(k_x, k_y) \, dk_x \, dk_y. \tag{3.5}$$

Each surface wavenumber component radiates as a single plane wave with vertical wavenumber $k_z = \sqrt{k_0^2 - k_x^2 - k_y^2}$. Where $k_x^2 + k_y^2 < k_0^2$ this is real—a genuine propagating wave—and the surface pressure is tied to the surface velocity by the half-space radiation impedance

$$\tilde{p} = Z_{\mathrm{r}}\, \tilde{v}, \qquad Z_{\mathrm{r}} = \frac{\rho_0 c_0\, k_0}{\sqrt{k_0^2 - k_x^2 - k_y^2}}. \tag{3.6}$$

Where $k_x^2 + k_y^2 > k_0^2$, k_z is imaginary, Z_{r} is purely reactive, and the component is evanescent: it stores energy at the surface but radiates none. So only the disk $k_x^2 + k_y^2 < k_0^2$—the *radiation circle*—contributes,

$$W = \frac{\rho_0 c_0}{2(2\pi)^2} \iint_{k_x^2+k_y^2<k_0^2} \frac{k_0}{\sqrt{k_0^2 - k_x^2 - k_y^2}}\, |\tilde{v}(k_x, k_y)|^2\, \mathrm{d}k_x\, \mathrm{d}k_y. \tag{3.7}$$

This is the physical heart of the subject: a surface radiates only through the parts of its wavenumber content that oscillate in space more slowly than sound—the "supersonic" components. Everything faster is trapped.

Now insert the finite panel. An infinite plate carrying a bending wave has velocity $v_\infty = v_0\, \mathrm{e}^{-\mathrm{i}(k_{px}x + k_{py}y)}$: a single wavenumber (k_{px}, k_{py}), an infinitely sharp spike in the wavenumber plane, which lies *outside* the radiation circle below coincidence and therefore radiates nothing. A real panel is that wave seen through a rectangular window, $v = v_\infty\, \Pi(x, y)$, and multiplying by a window is convolving in wavenumber:

$$\tilde{v}(k_x, k_y) = v_0\, \hat{\Pi}(k_x - k_{px},\, k_y - k_{py}), \qquad \hat{\Pi}(\kappa_x, \kappa_y) = L_x \operatorname{sinc}\frac{\kappa_x L_x}{2}\; L_y \operatorname{sinc}\frac{\kappa_y L_y}{2}, \tag{3.8}$$

with $\operatorname{sinc}(u) = \sin u/u$. The window smears the infinitely sharp spike into a pattern with skirts that *do* reach inside the radiation circle—and that leakage is exactly how a finite panel manages to radiate below coincidence. Squaring, $|\tilde{v}|^2 = |v_0|^2 L_x^2 L_y^2 \operatorname{sinc}^2\frac{(k_x - k_{px})L_x}{2} \operatorname{sinc}^2\frac{(k_y - k_{py})L_y}{2}$.

The radiation efficiency compares W with the power an ideal piston of the same mean-square velocity would emit, $\sigma = W/(\rho_0 c_0 S \langle v^2 \rangle)$, with $\langle v^2 \rangle = |v_0|^2/2$ and $S = L_x L_y$. Putting Eq. (3.7) in and cancelling common factors,

$$\sigma = \frac{L_x L_y}{(2\pi)^2} \iint_{\mathrm{disk}} \frac{k_0}{\sqrt{k_0^2 - k_x^2 - k_y^2}} \operatorname{sinc}^2\frac{(k_x - k_{px})L_x}{2} \operatorname{sinc}^2\frac{(k_y - k_{py})L_y}{2}\, \mathrm{d}k_x\, \mathrm{d}k_y. \tag{3.9}$$

Two cosmetic steps finish it. Use $\operatorname{sinc}^2 u = [1 - \cos 2u]/(2u^2)$ to rewrite each factor as $2W_x$, $2W_y$ with $W_x = [1 - \cos((k_x - k_{px})L_x)]/[(k_x - k_{px})L_x]^2$; the two factors of two combine with the $(2\pi)^2 = 4\pi^2$ to leave S/π^2. Then change to polar coordinates $k_x = k_r \cos\varphi$, $k_y = k_r \sin\varphi$, $\mathrm{d}k_x\, \mathrm{d}k_y = k_r\, \mathrm{d}k_r\, \mathrm{d}\varphi$. The result is exactly Eq. (3.2),

$$\sigma(k_p, \tau) = \frac{S}{\pi^2} \iint_{k_r<k_0} W_x\, W_y\, \frac{k_0\, k_r}{\sqrt{k_0^2 - k_r^2}}\, \mathrm{d}k_r\, \mathrm{d}\varphi. \tag{3.10}$$

The entire finite-size correction, then, is nothing more mysterious than an honest accounting of which wavenumbers a bounded panel is able to launch into the air.

Chapter 4

Thick plates

[Validated]

4.1 When thin-plate theory breaks

Recall the bending-wave speed implied by Chapter 2, $c_{\mathrm{B}} = \omega/k_{\mathrm{B}} = \sqrt{\omega}\,(D/m)^{1/4}$. It grows without limit as $\sqrt{f}$. For a thick, stiff element—concrete, masonry, laminated glass—this eventually predicts a bending wave *faster than the material's own shear wave*, which is physically impossible. Thin-plate theory is simply the wrong model there, and using it would misplace the coincidence dip and invent a high-frequency recovery that does not occur.

4.2 Shear, rotary inertia, and the Mindlin dispersion

Kirchhoff's thin plate makes two simplifications we must now undo. It assumes a line drawn normal to the plate stays both straight and perpendicular to the mid-surface as the plate bends—no transverse shear—and it ignores the rotational inertia of the plate's own thickness. Mindlin's theory keeps both, and to do so it needs *two* fields where Kirchhoff needed one: the transverse displacement $w(x,t)$ and the rotation of the cross-section $\psi(x,t)$. In thin-plate theory these are locked together, $\psi = \partial w/\partial x$; Mindlin lets them differ, and their difference is precisely the transverse shear strain,

$$\gamma = \frac{\partial w}{\partial x} - \psi. \tag{4.1}$$

For a straight-crested flexural wave (dependence on x only) the plate obeys two coupled equations. The first is the *transverse force balance*: the gradient of the shear force accelerates the mass,

$$\kappa^2 G h \left(\frac{\partial^2 w}{\partial x^2} - \frac{\partial \psi}{\partial x} \right) = \rho h \, \frac{\partial^2 w}{\partial t^2}. \tag{4.2}$$

The second is the *moment balance*: the gradient of the bending moment, less the couple carried by the shear force, drives the rotary inertia of the cross-section,

$$D \, \frac{\partial^2 \psi}{\partial x^2} + \kappa^2 G h \left(\frac{\partial w}{\partial x} - \psi \right) = \rho \, \frac{h^3}{12} \, \frac{\partial^2 \psi}{\partial t^2}. \tag{4.3}$$

Here $D = Eh^3/[12(1-\nu^2)]$ is the bending rigidity, $\kappa^2 G h$ (with $G = E/[2(1+\nu)]$ and $\kappa^2 \approx 5/6$) is the shear rigidity, ρh is the areal mass, and $\rho h^3/12$ is the rotary inertia per unit area. Write $S = \kappa^2 G h$, $m = \rho h$, $J = \rho h^3/12$ for brevity.

Now seek plane waves $w = W \, \mathrm{e}^{\mathrm{i}(kx - \omega t)}$ and $\psi = \Psi \, \mathrm{e}^{\mathrm{i}(kx - \omega t)}$. Each spatial derivative brings down $\mathrm{i}k$, each time derivative $-\mathrm{i}\omega$. Equation (4.2) becomes

$$(Sk^2 - m\omega^2) \, W + \mathrm{i}kS \, \Psi = 0, \tag{4.4}$$

and Eq. (4.3) becomes

$$\mathrm{i}kS\,W + (-Dk^2 - S + J\omega^2)\,\Psi = 0. \tag{4.5}$$

This is a homogeneous linear system for the amplitudes (W, Ψ); it has a non-trivial solution only if its determinant vanishes,

$$(Sk^2 - m\omega^2)(-Dk^2 - S + J\omega^2) - (\mathrm{i}kS)^2 = 0. \tag{4.6}$$

The cross term is $-(\mathrm{i}kS)^2 = +k^2S^2$. Multiplying out and collecting powers of k^2, the S^2k^2 pieces cancel and we are left with the *Mindlin dispersion relation*,

$$k^4 - \left(\frac{J}{D} + \frac{m}{S}\right)\omega^2\,k^2 + \frac{mJ}{SD}\,\omega^4 - \frac{m}{D}\,\omega^2 = 0. \tag{4.7}$$

It is a quadratic in k^2, whose physical (propagating flexural) root is

$$k_{\mathrm{eff}}^2 = \tfrac{1}{2}\left[\left(\tfrac{J}{D} + \tfrac{m}{S}\right)\omega^2 + \sqrt{\left(\tfrac{J}{D} - \tfrac{m}{S}\right)^2\omega^4 + \tfrac{4m}{D}\,\omega^2}\,\right], \tag{4.8}$$

which the engine solves directly. (The other root is the thickness-shear branch, evanescent in this band.)

Two limits confirm it behaves. At *low* frequency the ω^4 terms are negligible and Eq. (4.7) reduces to $k^4 = (m/D)\omega^2$, so $k_{\mathrm{eff}} \to (m\omega^2/D)^{1/4}$—exactly Kirchhoff's bending wave of Chapter 2. At *high* frequency the leading balance gives $k_{\mathrm{eff}} \to \omega\sqrt{m/S} = \omega\sqrt{\rho/(\kappa^2 G)} = \omega/c_s$: the flexural wave has become a shear wave and its speed *saturates* at $c_s = \sqrt{\kappa^2 G/\rho}$ rather than growing without bound as $\sqrt{f}$. Dropping rotary inertia ($J \to 0$) collapses Eq. (4.7) to the simpler $k^4 - (m/S)\omega^2k^2 - (m/D)\omega^2 = 0$; we retain the full form because for thick, stiff elements the J term is *not* negligible in the audible band—for 150 mm concrete it is comparable to the bending term by 8 kHz.

4.3 Worked example: 150 mm concrete versus 13 mm gypsum

Concrete ($\rho = 2300$ kg m^{-3}, $E = 30$ GPa, $\nu = 0.2$) has shear speed $c_s = 2128$ m s^{-1}. Thin-plate theory drives its bending speed to 2832 m s^{-1} at 8 kHz—faster than shear, which is nonsense. The Mindlin solution correctly saturates, climbing to about 1900 m s^{-1} and bending toward c_s. Meanwhile 13 mm gypsum is barely touched: $k_{\mathrm{eff}}/k_{\mathrm{thin}} = 1.01$ at 8 kHz. It is acoustically *thin* across the whole audible band, so the correction leaves it alone. That selectivity—repairing the thick element while not disturbing the thin one—is the signature of correct physics rather than a fudge. *[Engine module M3, dispersion limits verified.]*

4.4 What this does not yet capture

The shear-corrected dispersion gives the coincidence shift and the high-frequency bending-speed plateau, but not the discrete thickness (dilatational) resonances of a very thick element—standing compressional waves across the thickness. Those require the full solid-elastic transfer matrix, the masonry/concrete module of Chapter 7.

Chapter 5

The double leaf

[Validated]

5.1 Why two leaves beat one heavy one

The mass law of Chapter 2 is a hard bargain: 6 dB per doubling of mass. Doubling forever is not a design strategy—it is a structural engineer's nightmare. The escape is to stop fighting inertia with inertia and instead *decouple*: split the mass into two leaves separated by an air gap. Below a certain resonance the two leaves move together and you get only the single wall of their combined mass; but above it the air spring lets the second leaf lag the first, and the isolation grows far faster than any mass law. Every lightweight partition in the world is built on this one idea, so we had better derive that resonance carefully.

5.2 The air spring, and the mass–air–mass resonance

Trap a layer of air of thickness d between two large plates and squeeze it slowly. Compressing a gas column by δ changes its volume fraction by δ/d, and the pressure rises by $\Delta p = K\,\delta/d$, where K is the bulk modulus of air. For sound-speed compressions $K = \rho_0 c_0^2$, so the trapped layer behaves as a spring of stiffness per unit area

$$s_{\text{air}} = \frac{\rho_0 c_0^2}{d}. \tag{5.1}$$

Now hang two leaves of areal masses m_1 and m_2 on that spring and ask for the frequency at which they oscillate against each other. Two masses joined by one spring oscillate with the *reduced* mass $m_r = m_1 m_2/(m_1 + m_2)$ (each mass sees the spring, but the centre of mass stays put), so

$$\omega_0^2 = \frac{s_{\text{air}}}{m_r} = \frac{\rho_0 c_0^2\,(m_1 + m_2)}{d\,m_1 m_2}. \tag{5.2}$$

This is the *mass–air–mass* resonance, and it is the single most important number in double-wall design: below f_0 the wall is only its combined mass; at f_0 the two leaves resonate and the insulation *dips*; above f_0 the isolation climbs steeply until other physics (cavity resonances, coincidence, structure-borne bridges) intervenes.

Worked example. Two sheets of 13 mm plasterboard ($m_1 = m_2 = 10\ \text{kg}\,\text{m}^{-2}$) across a 90 mm cavity: $\rho_0 c_0^2 = 1.21 \times 343^2 \approx 1.42 \times 10^5$ Pa, so $\omega_0^2 = 1.42 \times 10^5 \times 20/(0.09 \times 100) = 3.16 \times 10^5$ and $f_0 \approx 90$ Hz. One refinement worth knowing: in a cavity well filled with porous material the compressions exchange heat with the fibres and become nearly *isothermal*, so K drops from γP_0 to P_0 and f_0 falls by $\sqrt{\gamma} \approx 1.18$—to about 76 Hz here. The fill does not just absorb; it retunes.

5.3 The cavity as a transmission line

The spring picture is honest only while the cavity is much thinner than a wavelength. In general the air layer carries waves, and we need its exact transfer matrix. Let the layer occupy $0 < z < d$, with normal velocity and pressure (p, v) on each face, oblique propagation with normal wavenumber $k_z = \sqrt{k_c^2 - k_t^2}$ (here k_c is the wavenumber *in the cavity medium* and k_t the conserved trace wavenumber). Write the field as a sum of standing waves, $p = A\cos k_z z + B\sin k_z z$; the momentum equation $\mathrm{i}\omega\rho\, v = -\partial p/\partial z$ then gives $v = (k_z/\mathrm{i}\omega\rho)\,(A\sin k_z z - B\cos k_z z)$. Evaluating both at $z = 0$ and $z = d$ and eliminating A, B yields the fluid-layer transfer matrix

$$\begin{pmatrix} p(0) \\ v(0) \end{pmatrix} = \begin{pmatrix} \cos k_z d & \mathrm{i}\,\dfrac{\omega\rho}{k_z}\sin k_z d \\ \mathrm{i}\,\dfrac{k_z}{\omega\rho}\sin k_z d & \cos k_z d \end{pmatrix} \begin{pmatrix} p(d) \\ v(d) \end{pmatrix}. \tag{5.3}$$

Two limits check it. As $k_z d \to 0$ the matrix tends to identity plus the air-spring term—the lumped spring of the previous section. And when $k_z d = n\pi$ the off-diagonal terms vanish: the cavity is a half-wavelength resonator and transmits as if it were not there. Those *thickness resonances* are the high-frequency ripples every deep cavity exhibits.

5.4 Filling the cavity: the equivalent fluid

An empty cavity is a resonant pipe; a filled one is a lossy line. Porous fill works because the air in its pores is dragged viscously against the fibres (parameterised by the *flow resistivity* σ, in $\mathrm{Pa\,s\,m^{-2}}$) and exchanges heat with them. Both effects are captured, without tracking the fibres themselves, by giving the cavity air a *complex* effective wavenumber and impedance—an "equivalent fluid." The engine implements Miki's empirical regression: with $X = f/\sigma$,

$$Z_c = \rho_0 c_0 \left[1 + 0.070\,X^{-0.632} - \mathrm{i}\,0.107\,X^{-0.632}\right], \tag{5.4}$$

$$k_c = \frac{\omega}{c_0}\left[1 + 0.109\,X^{-0.618} - \mathrm{i}\,0.160\,X^{-0.618}\right], \tag{5.5}$$

after Miki (1990), *J. Acoust. Soc. Jpn. (E)* **11**(1), 19–24. The imaginary parts are the losses; substituting k_c into Eq. (5.3) damps the mass–air–mass dip and kills the thickness resonances—which is precisely what measured curves of filled walls show. (The full poroelastic theory, in which the *frame* also carries waves, is Chapter 7's business; the equivalent fluid is its limp-frame limit and is accurate for ordinary batts.)

One honest subtlety the validation forced on us: an *empty* cavity is not lossless either—panel edges, leakage and viscous boundary effects dissipate energy. Following Vigran (2010), the engine represents an air-filled cavity as an equivalent fluid with a small effective resistivity in the range $\sigma_{\mathrm{eff}} = 10$–$100\ \mathrm{Pa\,s\,m^{-2}}$; across that entire published range the predictions for the empty-cavity test walls move by less than a decibel (Chapter 6), so nothing delicate hangs on the choice.

5.5 No new machinery

Here is the payoff of Chapter 1: the double leaf needs *no new theory at all.* Its transfer matrix is the ordered product

$$\mathbf{T}_{\mathrm{double}} = \mathbf{T}_{\mathrm{leaf},1}\,\mathbf{T}_{\mathrm{cavity}}\,\mathbf{T}_{\mathrm{leaf},2}, \tag{5.6}$$

fed into the master formula Eq. (1.13) and the finite-size machinery of Chapter 3. The engine's general assembler reproduces the analytic two-leaf result identically (to machine precision), and triple or asymmetric build-ups are just longer products. Reading a predicted curve is now anatomy: combined-mass law up to f_0 ($\approx$ 76–90 Hz above), the resonance dip, the steep rise, the ripple of any surviving cavity resonances, and the coincidence dip of each leaf at its own f_{c}. What the airborne product *cannot* contain is the path through the studs that hold the two leaves apart—and that, deliberately, is the whole of the next chapter.

Chapter 6

Structure-borne transmission

[Validated — steel single-frame, line model; point connections, wood and resilient channels in progress]

6.1 The plateau, and a failure worth understanding

Compute the airborne insulation of a filled steel-stud wall with Chapter 5's machinery and you will predict, at high frequency, a magnificent 90–100 dB. Measure the wall and you find 60–65. The energy is not going through the air at all: it runs down the studs. Any honest model must carry that second, *structure-borne* path.

The traditional shortcut (Sharp's, and its descendants) computes an independent stud transmission and adds it: $\tau_{\text{total}} = \tau_{\text{air}} + \tau_{\text{stud}}$. It seems innocent, and it fails in a specific, instructive way. A parallel path sets a *floor* under τ_{total}; if τ_{stud} knows nothing of the leaves' coincidence, then at f_{c}—where the airborne path plunges into its dip—the total cannot follow. Our own engine, run that way, clamped a measured 48 dB coincidence dip at a flat 63 dB. The physics being violated is simple: at coincidence the *receiving leaf itself* becomes a perfect radiator, and *every* path that ends by shaking that leaf—airborne and stud-borne alike—must dip with it. The two paths share a leaf; they are not parallel resistors. The cure is not a better τ_{stud}; it is to solve the leaves, the cavity and the connections *together*.

6.2 First attempt at togetherness: the smeared spring

Couple the two leaves by a uniformly distributed ("smeared") spring of impedance Z_s per unit area and solve the five simultaneous conditions of the problem—incident side, leaf 1 with the spring force, the cavity matrix, leaf 2 with the reaction, transmitted side:

$$\begin{gathered} p_{1f} = 2p_i - Z_a v_1, \quad p_{1f} - p_{1b} = Z_1 v_1 + Z_s\,(v_1 - v_2), \\ \begin{pmatrix} p_{1b} \\ v_1 \end{pmatrix} = \mathbf{T}_{\text{cav}} \begin{pmatrix} p_{2f} \\ v_2 \end{pmatrix}, \quad p_{2f} - p_{2b} = Z_2 v_2 - Z_s\,(v_1 - v_2), \qquad p_{2b} = Z_a v_2. \end{gathered} \tag{6.1}$$

Eliminating everything gives $\tau = |2Z_a/M|^2$ with a closed-form M; setting $Z_s = 0$ collapses it *exactly* onto the airborne product of Chapter 5, which the engine verifies to machine precision. And the coupled solve does what the parallel sum could not: at coincidence, *both* paths dip.

But the smeared spring tells its own lie. Spread uniformly, spring and leaves form a coherent mass–spring–mass oscillator, and the model grows a deep, spurious resonance near $\omega \approx \sqrt{s\,(1/m_1 + 1/m_2)}$ that no measured wall shows. Real studs are not a film of springs; they are *discrete lines at spacing b*. Discreteness is not a detail here—it is the physics.

6.3 What periodicity does to a wave

Let the studs run vertically (y) at $x = mb$. An incident trace wave $\mathrm{e}^{-\mathrm{i}k_x x}$ striking a b-periodic structure cannot respond with a single wavenumber, because the structure's response must repeat the drive's phase under translation by b (Floquet's condition). The most general such response is a comb of *space harmonics*,

$$u(x) = \sum_{n=-\infty}^{\infty} u_n \,\mathrm{e}^{-\mathrm{i}k_{x,n}x}, \qquad k_{x,n} = k_x + \frac{2\pi n}{b}, \tag{6.2}$$

with k_y conserved along the studs. The periodic array of line forces obeys the same rule, and one identity makes the whole method tractable. The force distribution $g(x) = \sum_m \delta(x - mb)\,\mathrm{e}^{-\mathrm{i}k_x mb}$ satisfies $g(x+b) = \mathrm{e}^{-\mathrm{i}k_x b} g(x)$, so $g\,\mathrm{e}^{+\mathrm{i}k_x x}$ is genuinely b-periodic and has a Fourier series whose every coefficient is $1/b$ (only the $m = 0$ delta lives in one cell). Hence

$$\sum_m \delta(x - mb)\,\mathrm{e}^{-\mathrm{i}k_x mb} = \frac{1}{b}\sum_n \mathrm{e}^{-\mathrm{i}k_{x,n}x} : \tag{6.3}$$

a line-force array of amplitude F drives *every* harmonic, equally, with strength F/b. Periodicity scatters the single incident wavenumber into the whole comb—and it is exactly this scattering that dissolves the smeared model's fictitious coherent resonance.

6.4 The per-harmonic system

Each harmonic n has its own trace wavenumber $k_{t,n} = \sqrt{k_{x,n}^2 + k_y^2}$ and, because the leaves and the cavity are laterally uniform, the harmonics couple *only* through the localized stud forces. Per harmonic, then, we need the dynamic stiffness (force per area per displacement) of each ingredient.

The leaf. From Chapter 2's impedance, $K_p = \mathrm{i}\omega Z_p = m\omega^2\big[(k_t/k_{\text{eff}})^4(1+\mathrm{i}\eta) - 1\big]$, with k_{eff} the thin-plate or Mindlin wavenumber as appropriate.

The outer air. A half-space driven at trace k_t responds with $p = \mathrm{i}\omega^2\rho_0 u/k_z$, $k_z = \sqrt{k_0^2 - k_t^2}$: for radiating harmonics ($k_t < k_0$) this is *radiation damping*; for evanescent ones $k_z = -\mathrm{i}\sqrt{k_t^2 - k_0^2}$ and the load is a pure added mass. Both cases fall out of the one formula with the right branch.

The cavity. Here is the neatest derivation in the chapter. Let the cavity fluid (effective density ρ_c, wavenumber k_c, so $k_z = \sqrt{k_c^2 - k_t^2}$) be bounded by faces moving with displacements u_1 at $z = 0$ and u_2 at $z = d$. Writing $p = A\cos k_z z + B\sin k_z z$ and imposing the two boundary velocities via $\mathrm{i}\omega\rho_c v = -\partial p/\partial z$ fixes A and B, and the face pressures come out as

$$p(0) = \frac{\omega^2\rho_c}{k_z}\Big[u_1\cot k_z d - u_2\csc k_z d\Big], \qquad p(d) = \frac{\omega^2\rho_c}{k_z}\Big[u_1\csc k_z d - u_2\cot k_z d\Big]. \tag{6.4}$$

The cot terms are what each leaf feels of its own cavity loading; the csc terms are the coupling through the gap. Assemble everything and the n-th harmonic obeys the 2×2 system

$$\begin{pmatrix} K_{11} & K_{12} \\ K_{12} & K_{22} \end{pmatrix}\begin{pmatrix} u_{1,n} \\ u_{2,n} \end{pmatrix} = \begin{pmatrix} 2P_0\,\delta_{n0} - F_1/b \\ +F_2/b \end{pmatrix}, \qquad \begin{aligned} K_{11} &= K_{p1} + \frac{\mathrm{i}\omega^2\rho_0}{k_{z,\text{air}}} + \frac{\omega^2\rho_c\cot k_{z,c}d}{k_{z,c}}, \\ K_{12} &= -\frac{\omega^2\rho_c\csc k_{z,c}d}{k_{z,c}}, \end{aligned} \tag{6.5}$$

(K_{22} mirrors K_{11} with leaf 2), following Legault & Atalla (2010). The incident sound drives only the specular harmonic $n = 0$ (with the blocked pressure $2P_0$); the stud forces F_1, F_2 drive them all.

6.5 The junction, and closing the loop

What is a stud, mechanically? Per unit length: a spring of stiffness s facing each leaf with a mass m' between—flange, body, flange. Solving that little three-element chain for the forces in terms of the two leaf velocities $v_i = \mathrm{i}\omega U_i$ at the stud line gives

$$F_1 = a_{11}v_1 - a_{12}v_2, \quad F_2 = a_{12}v_1 - a_{11}v_2, \qquad a_{11} = \frac{s}{\mathrm{i}\omega}\frac{s-\omega^2 m'}{2s-\omega^2 m'}, \quad a_{12} = \frac{s}{\mathrm{i}\omega}\frac{s}{2s-\omega^2 m'}. \tag{6.6}$$

With $m' = 0$ this is a plain spring of total line stiffness $K' = s/2$, transmitting $F = (K'/\mathrm{i}\omega)(v_1 - v_2)$; with mass, the junction isolates at high frequency like any mount. The loop closes through one physical statement: the stud feels the leaves *at its own line*, where the displacement is the sum over all harmonics, $U_i = \sum_n u_{i,n}$. Solving each harmonic of Eq. (6.5) by superposition (drive part plus force part), summing, and inserting into the junction law yields two linear equations for the two unknowns F_1/b, F_2/b—a 2×2 solve, and the whole periodic problem is done. Finally each harmonic radiates independently: a transmitted harmonic of amplitude $u_{2,n}$ carries intensity $\omega^3\rho_0|u_{2,n}|^2/(2k_{z,n})$ if it is supersonic, nothing if it is evanescent—except that on a *finite* panel Chapter 3's windowed radiation efficiency $\sigma(k_{t,n})$ replaces the sharp cutoff, and even subsonic harmonics leak. The engine applies the same verified σ to every harmonic.

Three exact checks pin the machinery down. With no studs the formulation reproduces the transfer-matrix result of Chapter 5 through entirely different algebra (agreement $\sim 10^{-13}$). Truncated to the single harmonic $n = 0$ it collapses *identically* onto the smeared model—which is therefore not discarded but contained, as the $N = 0$ limit. And the harmonic sum converges by $N \approx 16$ in the worst case. When it runs, the spurious smeared resonance is gone, the coincidence dip is native, and a massive junction isolates at high frequency, all as physics demands.

6.6 What the stud actually is: measurement

Everything above is exact mechanics of an *idealised* junction; the one thing theory cannot supply is the stiffness of a real, cold-formed steel C-stud, flanges flexing under screwed plasterboard. That number has been measured—by inverting sound-insulation theory against 126 laboratory steel-stud walls (Davy, Guigou-Carter & Villot, 2012) and extended to point models and resilient channels (Hirakawa & Davy, 2015). The engine uses the 2012 law: the equivalent translational compliance of the line connection is

$$C_M(f) = \min\Big(1.74\, f^{-1.81}\, m_r^{-1.40}\, b^{-0.75}\, d^{+0.28},\ \ 9.3\times10^{-5}\, m_r^{-1.09}\, d^{+0.80}\Big) \quad [\mathrm{Pa}^{-1}], \tag{6.7}$$

with f in Hz, the reduced mass $m_r = m_1m_2/(m_1+m_2)$ in $\mathrm{kg\,m^{-2}}$, and b, d in metres; the line stiffness fed to the junction is $K' = 1/C_M$ (and $m' = 0$, the mass being absorbed in the effective law). Note what the law says physically: below $\sim$500 Hz the compliance is constant, while above it the stud apparently *stiffens* with frequency—the leaves progressively fail to engage the flange's full flexibility as their bending wavelength shrinks. Two independent determinations (the 2015 study; Poblet-Puig et al.'s numerical characterisation, 2009) agree with Eq. (6.7) in the high-frequency range, which is the kind of agreement that lets one sleep.

6.7 Validation: the steel walls

Now the reckoning. Thirty-one insulated single-frame steel-stud walls from the NRCC laboratory series; the periodic line model; finite-size radiation on every harmonic; third-octave values computed as the band *power average* of τ (as the measurement standard defines them); and *every* input a published, measured anchor—Eq. (6.7) for the studs, the measured board moduli of Chapter 2, Miki's cavity fluid. No constant anywhere was adjusted to these walls.

Before the numbers, be precise about what is being compared, because the two figures below are not the same *kind* of figure. The reference value is Davy's model scored on this wall family *after* its one free function—the stud compliance—was obtained by inverting that very model against these very measurements: it is in-sample fit quality, the residual left when the loop is closed. Our figure is zero-adjustment *transfer*: the same compliance law, tuned to absorb another model's biases, transplanted into different physics with nothing re-fitted.

	This engine (transfer, no fitting)	Davy's model (in-sample fit)
overall mean	−0.1 dB	−0.1 dB
RMS of per-type std. dev.	3.4 dB (2.97 at $\eta = 0.04$)	2.4 dB

Let it be said plainly: parity with a calibrated simplified model is the *floor* this engine must clear, not its destination, and the table above does not yet clear it. What makes the result a foundation rather than a disappointment is its *decomposition.* The per-type means are essentially zero (±1.3 dB but one): the model tracks every construction correctly. The residual is a single systematic band signature shared by all types—−2.3 dB at 100 Hz, a +1–2 dB hump through 315–1250 Hz, and −5 to−7 dB at the coincidence dip—worth 2.1 dB RMS on its own; strike it out by physics and the expected spread is ≈ 2.7 dB. Nothing in the gap is scatter. Everything in it has a name:

The coincidence depth is damping-controlled, and here the published anchors genuinely conflict: the in-situ loss factor validated by the compliance studies is 0.03, while Davy's earlier anchor—validated against the same laboratory—is 0.04, which alone moves the spread from 3.4 to 2.97 dB. We refuse to pick the value that scores better on our own validation set (that would be fitting through the back door); until an independently *measured* in-situ damping settles it, $\eta \in [0.03, 0.04]$ is carried as a declared uncertainty, and the bands at and above f_c wear it as a confidence flag. One tested hypothesis is reported as a failure: band-centre sampling of the dip was suspected and the in-band average implemented—it changed almost nothing, so the dip is genuinely predicted too deep, not mis-sampled.

The mid-band hump is not private to this engine: Hirakawa & Davy's own comparison shows their calibrated line *and* point models rising above the experimental average in the same 160–630 Hz region. That residue lives in the shared, model-inverted junction law itself—which is precisely why the route past it is a junction characterised from the stud's *own mechanics* (finite-element and direct dynamic-stiffness measurement), not from inverting anyone's acoustic theory.

The 100 Hz tail belongs to the low-frequency modal regime this framework does not claim (Chapter 8's territory), compounded by the laboratory's own low-frequency limits.

Band statistics, however, are the physicist's metric; the consultant stakes a design on the single-number ratings, and the honest report must lead with them. Computed exactly per ISO 717-1 for the same thirty-one walls, the engine's R_w errs by −3.2 dB on average (−2.3 at $\eta = 0.04$), with only 6 (resp. 11) of 31 walls inside ±1 dB and a worst case of 10 dB. Why so much worse than the near-zero band bias? Because the rating's shifted-contour rule scores *deficiencies only*: eleven unbiased bands earn nothing back, while the few deep-deficit bands—the coincidence dip and the 100 Hz tail—drag the entire rating down. The same rule that exposes the problem locates the leverage: repair those specific bands and the rating error collapses disproportionately.

This forces the acceptance criteria to be stated once, plainly, and kept. A construction class is *finished* only when, with nothing fitted to the validation data, (i) predicted R_w and R_w+C_{tr} fall within ±1 dB of measurement for at least nine walls in ten, none worse than 2 dB, and (ii) the per-band residual is statistically indistinguishable from the *reproducibility of the measurements themselves*—for this dataset about 1.5–2 dB, evidenced by the 1.9 dB score of the mature airborne theory on the studless reference wall and by the several-decibel spread among nominally identical specimens. In one sentence: the prediction must be as trustworthy as building the wall and testing it; physics permits no claim beyond the data's own noise, and this book will make none—but nothing short of that ceiling counts as done. Against these criteria the steel class, at 6–11 of 31 within a decibel, is *not accepted*, and no benchmark-relative target ("beat the reference model") appears anywhere in this project again. (One side result stands unqualified: four further walls with *empty* cavities land at 3.8 dB using Vigran's published effective resistivity, and barely move across his entire stated range.)

Three lessons paid for this result and are worth stating plainly. Coupling must live *inside* the solve, or coincidence is clamped. Discreteness (periodicity) is physics, not detail, or a phantom resonance appears. And when a fitted constant disagrees with a measured one—as the old critical-frequency product disagreed with the measured board moduli—the measurement wins; correcting that single anchor removed a +11 to +14 dB shoulder at one stroke.

6.8 What remains open

Real boards are attached by *screws*, and above the frequency where half a bending wavelength fits between them (∼300 Hz for 13 mm board at 300 mm centres) a line is the wrong idealisation: the connection is a two-dimensional *point lattice*, whose treatment doubles the harmonic expansion and takes the point-form compliances (units $\mathrm{m\,N^{-1}}$) of Hirakawa & Davy. Beyond topology, the constitutive law itself must graduate from model-inversion to *mechanics*: the stud's dynamic stiffness computed from its own cross-section (finite elements) and taken from direct dynamic-stiffness measurement, so that no part of the engine's accuracy rests on a constant fitted through somebody else's theory. Alongside sit an independently measured in-situ damping anchor to close the $\eta \in [0.03, 0.04]$ provenance band, the shared head- and base-plate path of staggered-stud walls, and the double-frame class. That is the work in progress that keeps this chapter's status tag honest—and it is the specified route from parity to superiority.

Chapter 7

The layer library

[**Planned**] Orthotropic and frequency-dependent stiffness (laminated-glass shear, cross-laminated timber, sandwich cores); solid elastic layers with thickness resonances (masonry, concrete); poroelastic Biot theory and its equivalent-fluid limit.

Chapter 8

The low-frequency regime

[Planned] Where the diffuse and infinite assumptions fail, the modal sound field, and the coupled vibro-acoustic wave-based solve that replaces the transfer matrix below the crossover (∼50–150 Hz, up to ∼500 Hz for lightweight double-leaf).

Chapter 9

Auralisation

[**Planned**] From the predicted TL spectrum to an audible rendering: a source signal (road traffic—the C_{tr} spectrum—or speech, or laughter), filtered by the build-up, convolved with the receiving room, and rendered binaurally.

Chapter 10

Worked problems

[**Planned**] Complete build-ups solved end to end, by hand alongside the engine, so every step can be followed.